Glucocorticoid-responsive genes in airway smooth muscle: a re-analysis of public RNA-seq data with DESeq2

Antoine Haddad

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1 Summary

This report re-analyses a public bulk RNA-seq experiment: four primary human airway smooth muscle cell lines, each treated with the glucocorticoid dexamethasone or left untreated. The original study (Himes et al., 2014) reported 316 differentially expressed genes. Using DESeq2 (Love et al., 2014) with a paired design, I find **4,081 genes** at an adjusted $p < 0.05$ (2,190 up, 1,891 down), and 951 with an additional $|\log_2 \text{fold change}| > 1$. All four genes the original paper named as well-known glucocorticoid targets (*DUSP1*, *KLF15*, *PER1*, *TSC22D3*), and *CRISPLD2* (the paper's headline gene), are significantly up-regulated here. Comparing with the original paper's supplementary list shows its 316 genes are almost all among mine (301 of the 304 I could test, all in the same direction), so the larger count is mostly additional genes with smaller effects (Discussion). Pairing samples by cell line (accounting for donor) changes the answer materially: an unpaired model finds 2,773 genes, of which 2,674 are also found by the paired model.

2 Background and question

Glucocorticoids are a mainstay of asthma treatment, but how they act on airway smooth muscle is poorly understood. Himes et al. (2014) profiled the transcriptome of four primary human airway smooth muscle cell lines with and without 18 hours of 1 µM dexamethasone. I ask three questions of the public data:

1. Which genes respond to dexamethasone, and do the paper's named genes appear among them?
2. How much does it matter that the same four cell lines were measured in both conditions?
3. What biology do the responding genes point to?

3 Data

The counts come from the Bioconductor `airway` package, which packages the public GEO series GSE52778 as a gene-level count table (Ensembl annotation, 8 samples, 63,677 genes). Design: 4 cell lines (N61311, N052611, N080611, N061011) × {untreated, dexamethasone}. Library sizes range from roughly 15 to 31 million reads per sample (Figure 1A). No data was generated for this report.

4 Methods

Filtering. Genes were kept if at least 10 reads were observed in at least 4 samples (16,139 of 63,677 genes kept). Four is the size of the smallest condition group.

Differential expression. DESeq2 negative-binomial GLM with design `~ cell + dex`, so each cell line serves as its own control. Wald test for the dexamethasone coefficient, Benjamini-Hochberg adjusted p-values (DESeq2 default with independent filtering), significance at adjusted $p < 0.05$. Log fold changes for plotting and ranking were shrunk with `apeglm` (Zhu et al., 2019). As a comparison, the same data were fitted with `~ dex` (no cell-line term).

Exploration. Variance-stabilising transformation (VST) for PCA and the heatmap.

Enrichment. Over-representation analysis of GO Biological Process terms (`clusterProfiler`) for genes with adjusted $p < 0.05$ and $|\log_2 \mathrm{FC}| > 1$, against the filtered genes as the background universe. Terms with adjusted $p < 0.05$ are reported.

Software. R 4.5.3, DESeq2 1.50.2, Bioconductor `airway` 1.30.0.

5 Results

5.1 Sample quality and structure

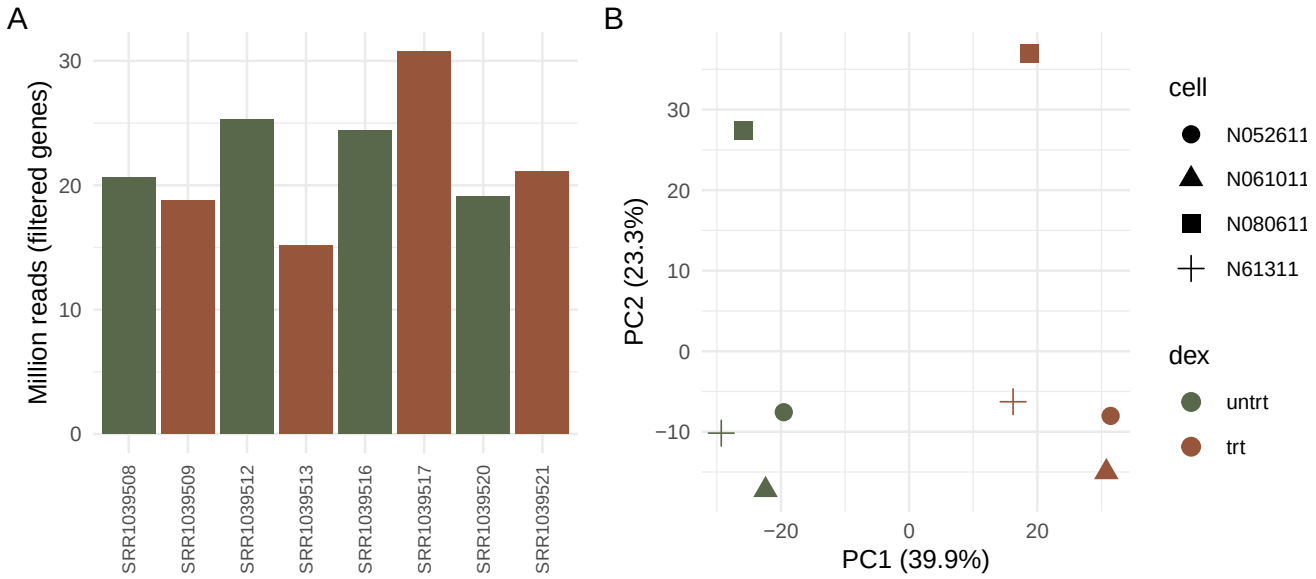


Figure 1: Library sizes and sample structure. (A) Total reads per sample. (B) PCA of VST-transformed counts: PC1 separates dexamethasone-treated from untreated samples; the cell lines separate along PC2 and PC3.

The dominant axis of variation is the drug: PC1 (39.9% of variance) separates treated from untreated samples ($R^2 = 0.95$ against treatment), while PC2 and PC3 together (40.1%) separate the four cell lines ($R^2 = 0.98$ and 0.96 against cell line). Treatment is a strong, consistent shift, but a large share of the remaining variance is between-line, which is the reason to model cell line explicitly.

5.2 Differential expression

At adjusted $p < 0.05$, 4,081 genes respond to dexamethasone (2,190 up, 1,891 down); 951 of these also have $|\log_2 \mathrm{FC}| > 1$ (Figure 2).

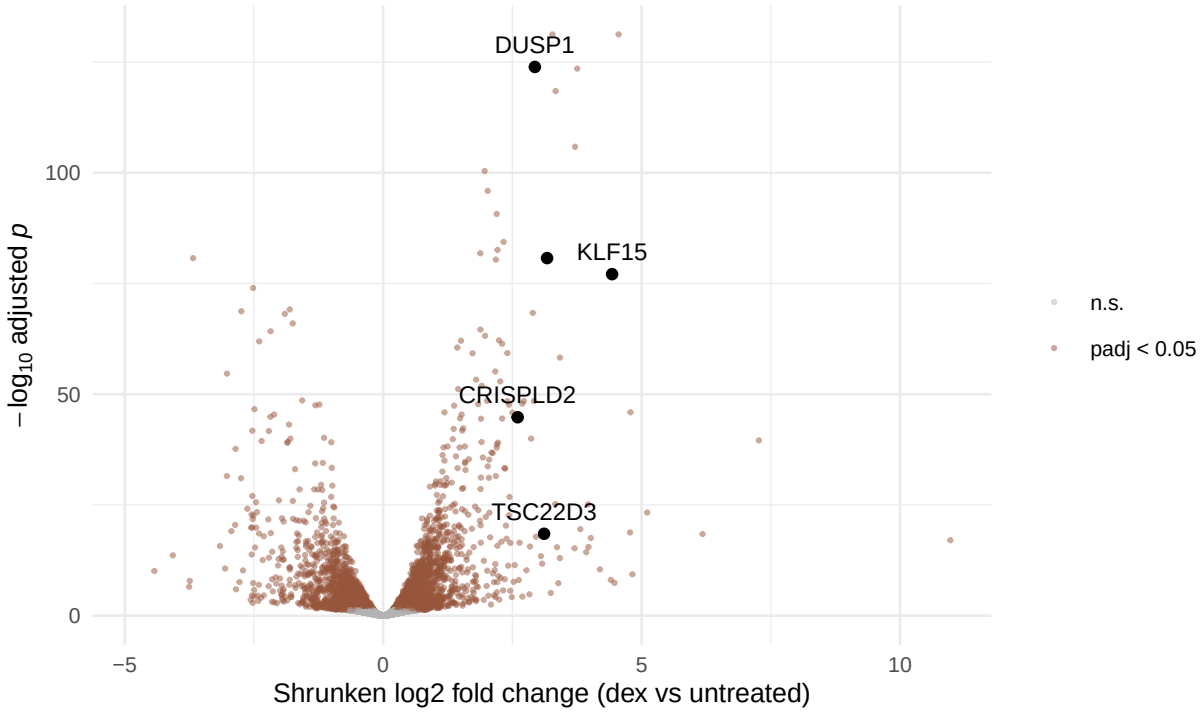


Figure 2: Volcano plot of shrunken log2 fold change against significance. Highlighted genes were named in the original paper.

Table 1: Genes highlighted by the original paper. All five are significantly up-regulated.

Gene	log2FC (shrunken)	Adjusted p
CRISPLD2	2.60	1.6e-45
DUSP1	2.94	1.3e-124
KLF15	4.43	7.8e-78
PER1	3.17	1.9e-81
TSC22D3	3.11	3.2e-19

Table 2: Ten most significant genes.

Gene	Mean count	log2FC (shrunken)	Adjusted p
SPARCL1	998	4.56	5.6e-132
CACNB2	496	3.28	5.6e-132
DUSP1	3412	2.94	1.3e-124
SAMHD1	12713	3.75	3.0e-124
MAOA	2344	3.34	3.6e-119
GPX3	12298	3.71	1.4e-106
STEAP2	3013	1.96	4.0e-101
NEXN	5399	2.02	1.2e-96
MT2A	3660	2.20	2.0e-91
ADAMTS1	30346	2.33	4.1e-85

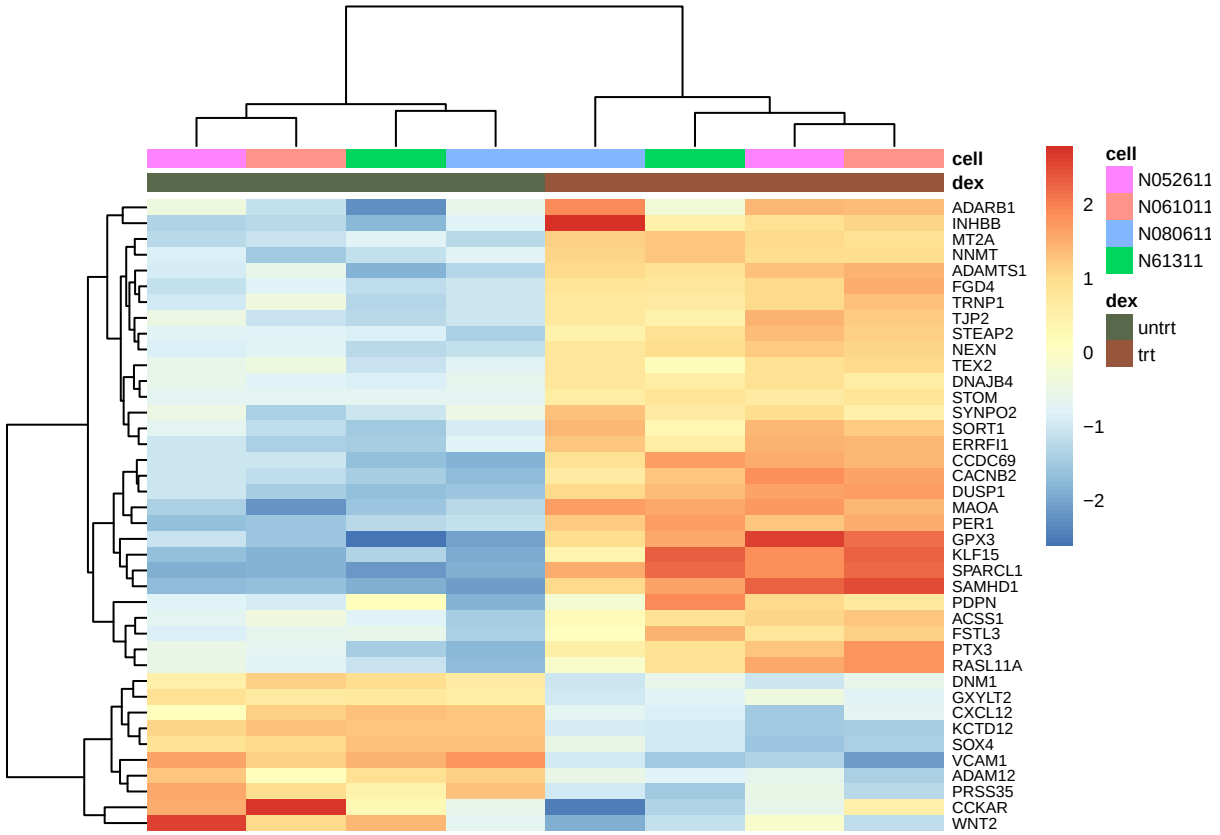


Figure 3: Top 40 genes by adjusted p-value, VST expression centred per gene. Hierarchical clustering of samples on these genes splits them into exactly the treated and untreated groups.

5.3 Does pairing by cell line matter?

An unpaired model (`~ dex`) finds 2,773 genes at the same threshold, against 4,081 for the paired model. Of the unpaired hits, 2,674 (96%) are also found by the paired model. Putting cell line in the design removes between-line variance from the error term, which is why the paired model has more power here. Ignoring the pairing would have missed about 1,308 genes at this sample size.

5.4 Biological themes

The top terms below are broad developmental and vascular/muscle-system processes. The axon-guidance terms are most likely driven by shared guidance-cue genes that also act in other tissues, so I would not read them as neuronal activity in these cells.

Table 3: Top GO Biological Process terms among genes with adjusted $p < 0.05$ and $|\log2FC| > 1$.

Term	Gene ratio	Adjusted p
axon guidance	38/789	9.1e-09
neuron projection guidance	38/789	9.1e-09
regulation of system process	61/789	1.1e-08
circulatory system process	64/789	8.0e-08
blood circulation	56/789	2.7e-07
axon development	59/789	2.9e-07
muscle system process	51/789	2.9e-07
plasma membrane bounded cell projection morphogenesis	70/789	4.0e-07
angiogenesis	63/789	4.0e-07

Term	Gene ratio	Adjusted p
blood vessel morphogenesis	69/789	4.0e-07

6 Discussion and limitations

Why 4,000 genes and not 316? I compared my result with the paper's Table S3 (the 316 genes, from Cuffdiff; supplementary file of a CC BY paper). Of those 316 genes, 304 are in my tested set (the other 12 fell below my expression filter or did not match by gene symbol); 301 of those 304 are significant here, and all 304 change in the same direction. The paper's list is therefore nearly a subset of mine. The additional 3,780 genes have much smaller effects: the median absolute $\log_2$ fold change is 0.58 across all my significant genes, against 1.73 for the paper's list (computed from its FPKM values). Restricting mine to $|\log_2 \text{FC}| > 1$ leaves 951 genes, of which 299 are in the paper's list. This is the pattern a less powerful test on the same four pairs would give, but I did not run Cuffdiff, so I cannot say the tool is the cause; the original filtering and the reference annotation are also untested. One detail worth knowing: Table S3's column headed "Ln[Fold Change]" matches $\log_2$ of the FPKM ratio on 95% of rows, so it appears to be $\log_2$ despite its label.

Sample size. Four cell lines is a small number of biological replicates. The paired design is the right use of that data, but results describe these four lines, not airway smooth muscle in general. Dispersion estimates from four pairs are themselves noisy.

Statistical significance is not effect size. With thousands of significant genes, adjusted p alone is a weak filter for prioritising follow-up. I use shrunken fold changes for ranking and the $|\log_2 \text{FC}| > 1$ subset for enrichment for that reason; the threshold is conventional rather than derived.

Enrichment. GO over-representation depends on the background universe and on the arbitrary fold-change cut. It is a hypothesis generator, not evidence of mechanism.

No validation. This is a computational re-analysis. The original study validated *CRISPLD2* by qPCR and knockdown; nothing here substitutes for that.

7 Provenance and reproducibility

All data are public (GEO GSE52778 via the Bioconductor `airway` package). The analysis code is this document. It was written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding every result before relying on it. Source and rendering instructions: `report/report.qmd`, rendered with `quarto render`. No part of this report derives from university coursework or assessed material.

References

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